

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Surprise Test - 2
 Duration: 10 minutes

Semester: Spring 2025
 Full Marks: 05

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section: <u>2</u>
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1. The following table represents the contents of the Data Memory:

Location	Data	Location	Data
1000	1001 0001	1008	1011 0001
1001	0010 0101	1009	0010 0100
1002	1001 0001	1010	1001 0000
1003	1000 1000	1011	1000 1000
1004	1011 1010	1012	1011 1011
1005	0100 0001	1013	0100 0001
1006	1000 0000	1014	1000 0000
1007	1110 0001	1015	1110 0000

→ LSByte

→ MSByte

So, forming 32 bit Data

1000 1000 1001 0000
 0010 0100 1011 0001

Note: The data memory above follows the same endianness as RISC-V
 little endian

Given the instruction:

LW $X_{19}, 8(X_{12})$

and register $X_{12} = 1000$

Determine the value stored in register X_{19} after executing the instruction.

Physical location = $1000 + 8$
 $= 1008$; Starting location

32
bit

Answer:

$X_{19} =$

32 bits

1011 0001 ✓

1111 1111 1000 1000 1001 0000 0010 0100

M

LW → Load word → performs signed extension.

MSBit of Data is 1. So rest of the 32 bits in the register will be filled by 1.

2. Construct the equivalent RISC-V code of the following high-level code.

if ($a > b$ and $\text{save}[i] \leq 5$):
 $c = 5$

means both conditions must satisfy.

The base address of a halfword array save is in register X_{19} . Also consider a, b, c, d and i are in registers $X_{20}, X_{21}, X_{22}, X_{23}, X_{24}$ respectively.

BLT $X_{21}, X_{20}, \text{If1}$

BEG X_0, X_0, Exit

If1:

Addi $X_5, X_0, 5$; $X_5 = 5$

Slli $X_6, X_{24}, 1$; $X_6 = \text{offset} = i \times 2^1$

Add X_6, X_{19}, X_6 ; $X_6 = \text{Base} + \text{offset}$

LH $X_7, 0(X_6)$; $X_7 = \text{save}[i]$

BGE $X_5, X_7, \text{If2}$

BEG X_0, X_0, Exit

If2:

Addi $X_{22}, X_0, 5$

Exit:

$$a > b = \frac{b < a}{X_{21} \quad X_{20}}$$

$$\text{save}[i] \leq 5 = 5 \Rightarrow \text{save}[i]$$